Experiment 11: Project LAST EXIT - Intelligent Disaster Evacuation System

Aim

To develop an intelligent system that coordinates rescue teams and helps people evacuate safely during a disaster.

Exercises:

1. **Knowledge Representation:** Represent rescue teams, affected areas, shelters, and their relationships using a suitable knowledge representation technique.
2. **Rule-Based Reasoning:** Create IF–THEN rules using forward or backward chaining to identify evacuation priorities and suggest suitable rescue actions.
3. **Handling Uncertainty:** Use fuzzy logic or probabilistic reasoning to make decisions when flood levels, road conditions, or emergency reports are uncertain.
4. **Resource Allocation:** Assign rescue teams and shelters while considering available resources, shelter capacity, and road conditions.
5. **Multi-Agent Coordination:** Use PGP or CNP to coordinate rescue teams, assign tasks, and resolve conflicts when multiple teams need the same resources.
6. **Simulation and Evaluation:** Test the system under different disaster scenarios. Introduce an unexpected event, such as a blocked evacuation route or an unavailable shelter, and observe how the system adapts. Visualize the results and compare evacuation time, resource usage, and the number of people evacuated safely.